# Potion Commotion!

By Group 15

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# Game Overview

## Game Concept

“A game where you will have to sell potions to customers while your workspace is CONSTANTLY betraying you!”

Potion Commotion takes a more traditional rapid-cooking game and brings added chaos through hazard, microgames to make sure the player has anything but a regular brewing experience. As shifts get more completed, things break more. Try and be able to pay off rent for as long as possible!

## Genre

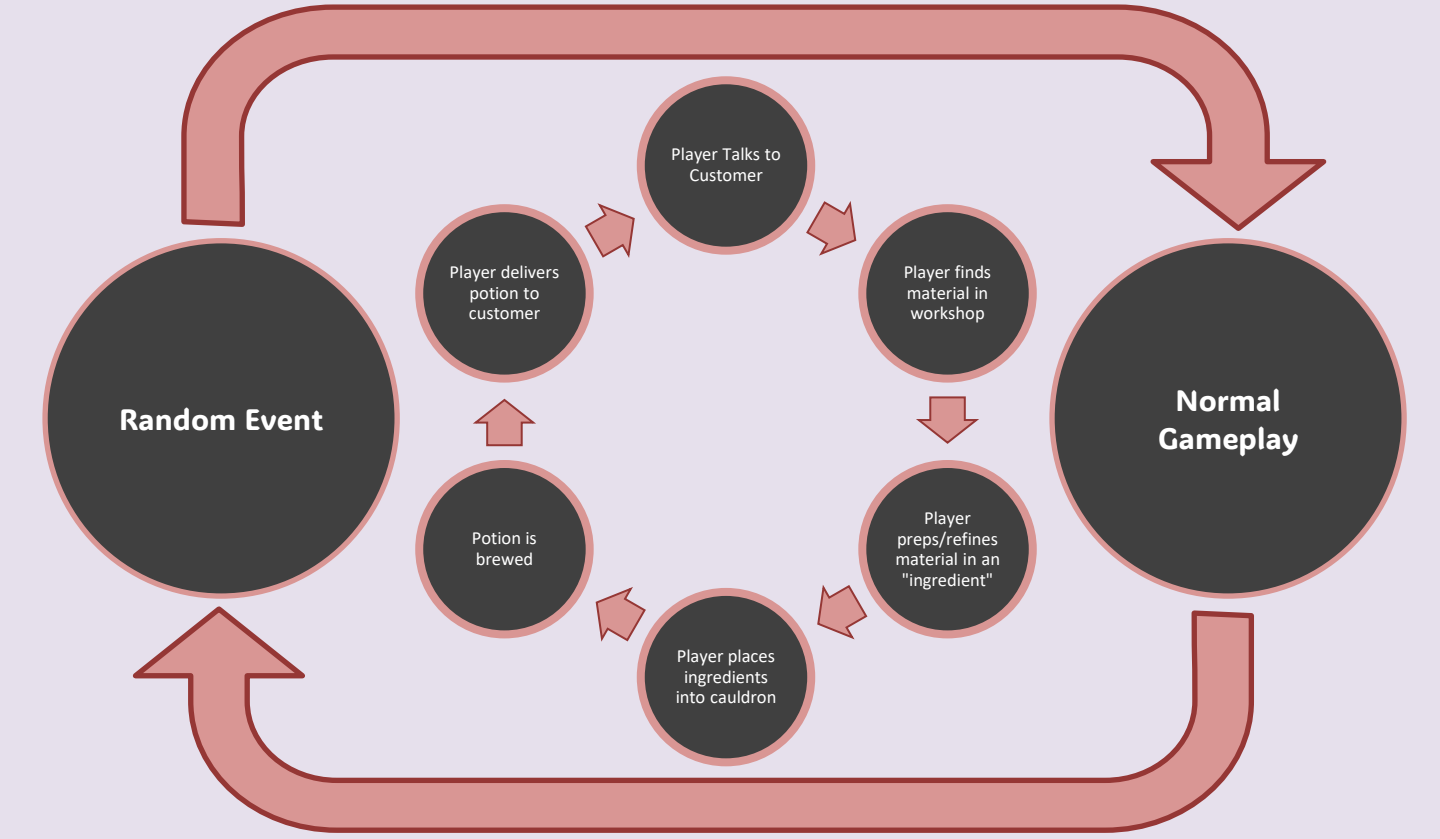
Crafting, singleplayer, simulation

## Target Audience

Potion Commotion is aimed to be E for everyone. There is very little violence, and the overall visual style is playful and light-hearted. There is no disturbing language or imagery however the chopping up of certain ingredients may be disturbing to some.

# Game Flow Summary

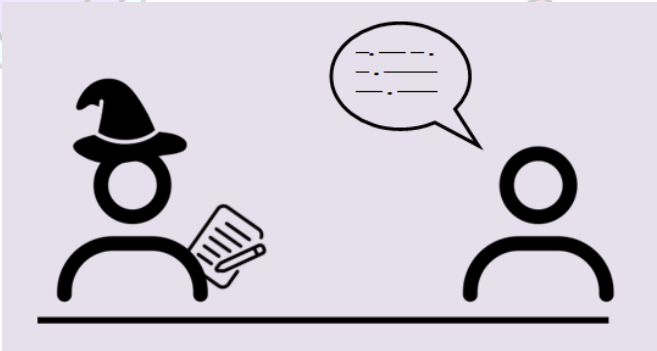
The game bounces between “Normal Gameplay” and “Random Events”. Normal Gameplay involves brewing and selling potions to customers (the main gameplay loop that the player is constantly working on). Random Events can occur at any time and throw off the loop. Through this, the player can resolve the Random Event to streamline the Normal Gameplay flow as much as possible.




# Normal Gameplay Flow

## *Player Talks to Customer*

The player interacts with a customer who responds with a potion they want. From here, the player looks up the ingredients needed to make the potions in the recipe book.



## *Player Finds Ingredients in Workshop*

Throughout the room, different types of ingredients are sorted into containers. The player must find the ingredients they need to interact with it and pick them up. The player can only need to repeat this step throughout the loop as they can only pick up one ingredient at a time.

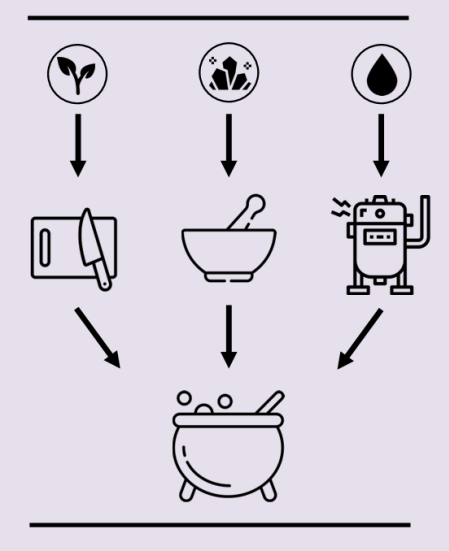


## *Player preps / refines ingredients*

There are many different crafting stations in the room the player can interact with. If they are holding the right ingredient, the can “prep” the ingredient by completing a short minigame. A few examples of minigames include:

- Solving Mini Logic Puzzles
- Mashing a button
- Using the mouse to chop or slice

These minigames take no longer than 15 seconds to complete however some of them require waiting and can be done adjacent to others.



## *Player places ingredients into Cauldron*

As ingredients are prepped, they can be placed into the cauldron in the centre of the room. The player can only hold one ingredient at a time so the player will be depositing ingredients into the cauldron throughout.

### *Potion is Brewed Cauldron*



As ingredients are prepped, they can be placed into the cauldron in the centre of the room. The player can only hold one ingredient at a time so the player will be depositing ingredients into the cauldron throughout.

### *Player Delivers Potion to Customer*

The player can scoop up the liquid in the cauldron and deliver it to the customer, receiving money in return. They can accomplish this by grabbing an empty bottle and bringing it to the cauldron.

## Look and Feel

The game is set in the standard medieval setting with a cuter approach akin to cosy games such as Animal Crossing. The overall tone will be very light-hearted to contrast with the darker story and panic of the game!

The visuals take on a more vibrant and hand-painted texture style, with brighter visuals and lighting to reflect a nighttime store. The models are made to be low poly, and most assets will have a slight “wonkiness” to show the run-down nature of the kitchen. Characters have a cute, round shape and many different hairstyles/clothing that will be swapped around.

The mood will balance the joyful colours and textures of light-hearted party games with the nighttime witchcraft aesthetic to make a joyful but uniquely mysterious look.

# Core Pillars of Design

## Chaos

All the game's mechanics, visuals and general game flow are built to create as much chaos for the player as possible. As time goes on, different stations and hazards will be introduced to blindside players and throw off the standard flow of the gameplay. Furthermore, the act of making potions and the effects are also chaotic

## Crafting

The core loop and gameplay system revolve around preparing ingredients, crafting potions and appeasing customers. The multiple crafting stations and combinations create a moderately complex crafting system in which the player must navigate and master to be the most time-efficient.

## Hazards

Hazards are placed to disrupt the flow of the game as much as possible and instil the level of chaos further! Every disruption can be fixed at the cost of time leading to less time for crafting and thus fewer potions.

## Time

The player is constantly against the clock to produce as many potions as they can in the shift. Time is of the essence as every ingredient/interruption will take time to complete and affect your income.

# Gameplay Systems & Mechanics

## Potion Brewing

### Overview

Potions are brewed in a cauldron with a minimum of two ingredients. Ingredients are created by taking a material and using it with a craftstation. The functionality of each craft station is outlined in its respective section. Each potion has a base cost of about 20 coins

### Materials

Ingredients come in four categories, each category contains three materials and can be used with certain craft stations

Plants

Plants can be used with the Chopping Board, the Mortar&Pestle, and the Boiler. They're the first Material introduced, as they're not conceptually complex

Creatures

Creatures can be used with the Chopping Board, the Furnace, and the Incubator. They're introduced after the Plants, as they're simple to understand, but still have some magical spice to fit with the theme

Bottled

Bottled can be used with the Boiler, the Enchanter, and the Incubator. These are introduced third, as they utilise some of the more complex craftstations

Minerals

Minerals can be used with the Mortar&Pestle, the Furnace, and the Enchanter. These are introduced last, and add a lot of overlapping functionality to some craftstations.

### Star Rating

After being prepared, ingredients are given a rating from 1-5 stars based on how well the craft station minigame was completed. After all the ingredients have been combined into a single potion, the star rating of the potion is calculated as the average of the ingredients star ratings rounded up to a whole number. When a potion is given to a customer, they pay an amount according to the star rating of the potion.

| Potion | Payment | Response |
|---|---|---|
| 5 Stars | 120% of base cost | Extra Happy |
| 4 Stars | 100% of base cost | Positive |
| 3 Stars | 80% of base cost | Neutral |
| 2 Stars | 50% of base cost | Tastes funny |
| 1 Star | 30% of base cost | Grumble about quality |
| Wrong | 0% of base cost | Complain that it's the wrong potion |

# Recipes

| | | | Ingredient 1 | | Ingredient 2 | | Ingredient 3 | |
|---|---|---|---|---|---|---|---|---|
| Day Introduced | Potion | Description | Craftstation | Material | Craftstation | Material | Craftstation | Material |
| 1 | Sleep | Causes slumber upon contact with skin. Over use will cause incurable coma. | Chopped | Mushroom | Ground | Curved | | |
| 2 | Health | Remedies wounds and cures sicknesses when consumed | Chopped | Curved | Burned | Scarab | | |
| | Love | Highly unethical, only use for pranks and ironic punishments | Ground | Lips | Burned | Lizard | | |
| 3 | Poison | Tastes like rotten lemon and dead rats | Ground | Mushroom | Chopped | Shark | | |
| 4 | Night Vision | | Ground | Curved | Boiled | Eyeballs | | |
| | Scale Skin | | Chopped | Lizard | Boiled | Gremlin | | |
| 5 | Levitation | Causes flight and a desire to eat seeds and worms | Chopped | Scarab | Boiled | Mushroom | | |
| | Greater Health | Restores lost limbs. Has a subtle cherry flavour | Chopped | Curved | Burned | Scarab | Boiled | Fish Bones |
| 6 | Truth | May cause spontaneous combustion of the underwear | Enchanted | Eyeball | Boiled | Lips | | |
| | Water Breath | | Enchanted | Fish Bones | Chopped | Shark | | |
| 7 | Growth | Side effect of mild boost in confidence and attractiveness | Enchanted | Gremlin | Ground | Mushroom | | |
| | Greater Poison | Causes immediate death upon contact with skin | Ground | Mushroom | Chopped | Shark | Enchanted | Eyeball |
| 8 | Blood Moon | Elixir of the damned, curses of the unholy denizens of hell. Goes great with pancakes | Burned | Stardust | Boiled | Gremlin | Incubated | Lizard |
| | Strength | | Ground | Crystal | Burned | Shark | | |
| 9 | Charm | | Incubated | Fish Bones | Chopped | Lips | | |
| | Foresight | | Incubated | Eyeball | Burned | Lizard | | |
| | Blindness | Causes True Sight when combined with potion of Night Vision | Incubated | Scarab | Ground | Ammonite | | |
| 10 | Resurection | Draws souls back from beyond the veil. Ineffective on living beings | Enchanted | Ammonite | Incubated | Gremlin | | |
| | Stupidity | [No description, just scribbles] | Burned | Crystal | Enchanted | Fish Bones | | |
| | Greater Levitation | Do not consume outdoors | Chopped | Scarab | Boiled | Mushroom | Enchanted | Stardust |

# Craft Stations

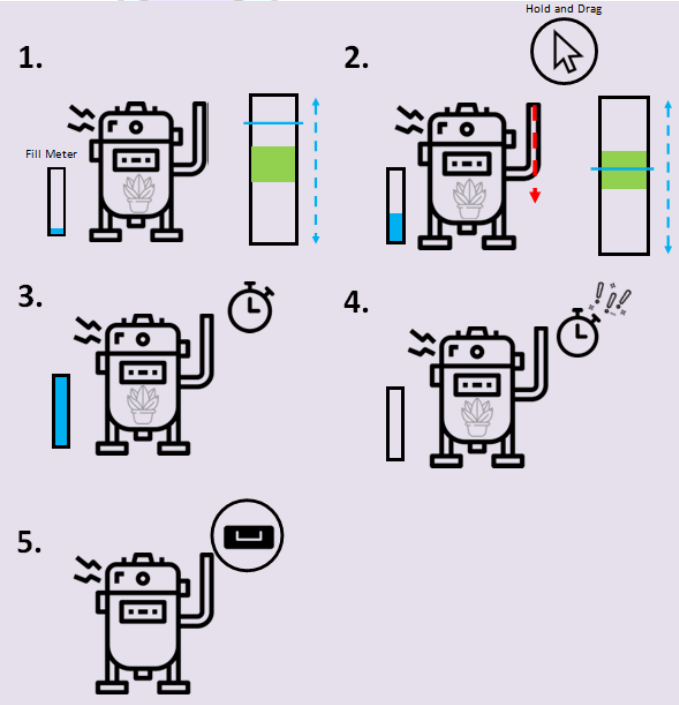
## Overview

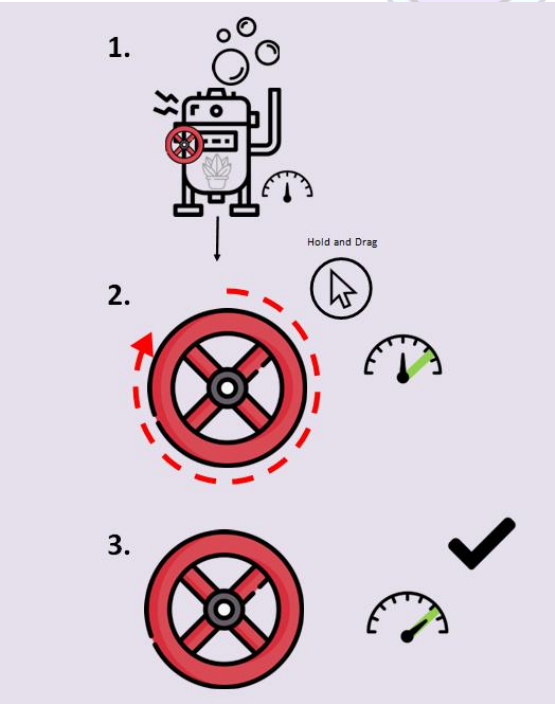
Craft stations are…

Things in red are ideas currently not in the game but may be added later in polish!

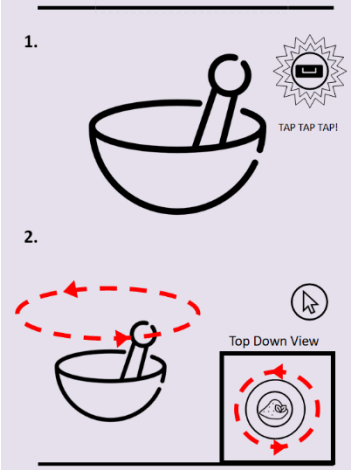
## Boiler

- **Function:**
  Place materials into the hatch and begin pumping by dragging your mouse down on the lever to pump and generate steam, causing the contents to boil. It initially starts slow and hard to pump but as you begin doing it, it becomes easier and lighter to pump faster. Ingredients like plants will transform into a water bottle with the plant inside. Ingredients prepared here are considered “Boiled”
- **Malfunction:**
  If the Boiler malfunctions, it will bubble over, spilling water and making the floor slippery, which reduces movement speed. To repair it, adjust the pressure by turning the valve until the pressure gauge is in the green zone and pressing spacebar.



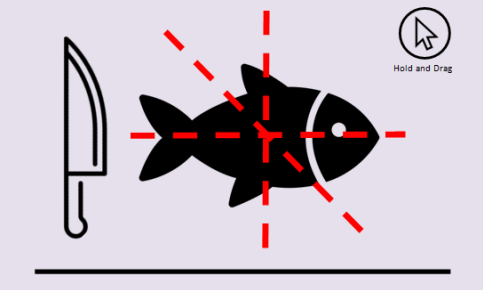



## Mortar & Pestle




- **Function:**
  Press the spacebar to grind the ingredients in the mortar. The faster you press, the quicker the grinding process. Ingredients prepared here are considered “Grinded.”
- **Malfunction:**
  None.

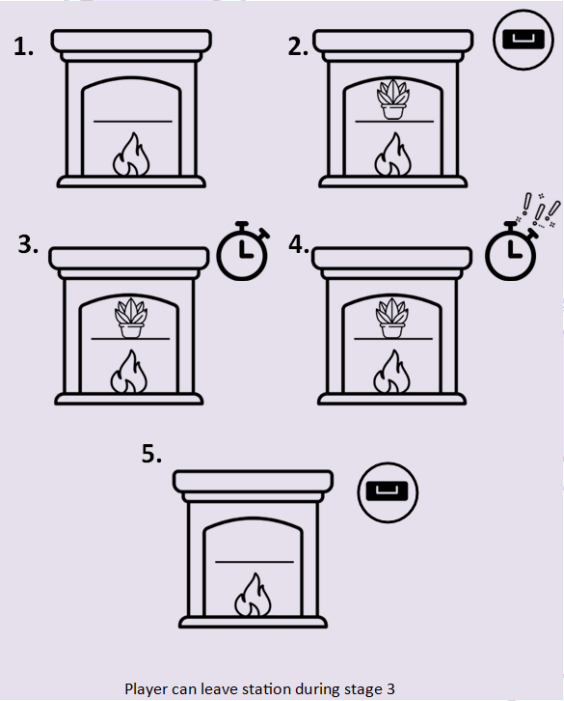
## Chopping Board

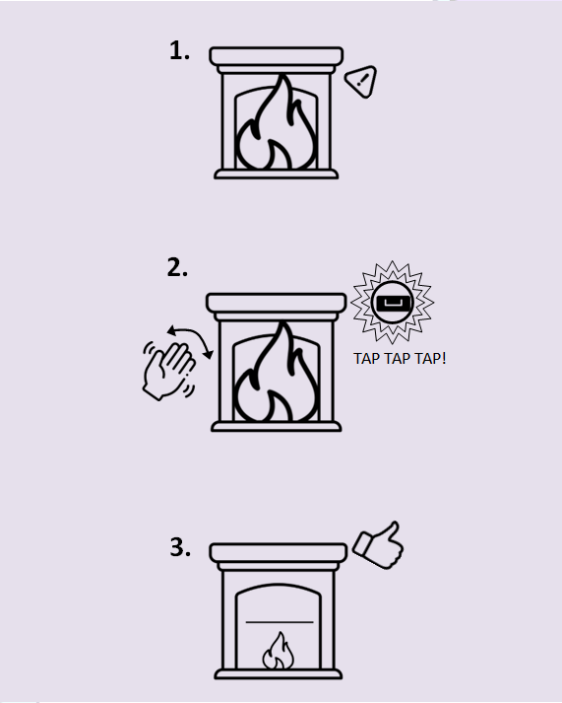



- **Function:**
  Ingredients will have different line patterns that must be followed. Use your mouse to drag the knife along these lines. The accuracy of your cuts will affect the quality of the ingredients. Ingredients prepared here are considered “Chopped.”
- **Malfunction:**
  None.

# Furnace

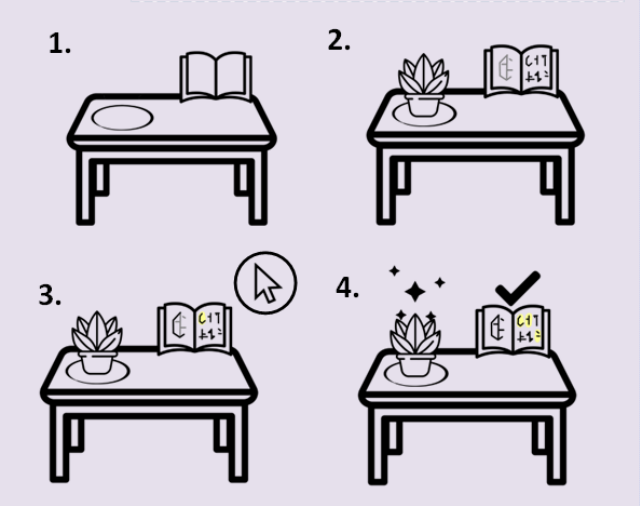
- **Function:**
  Place an ingredient on a plate and insert it into the furnace to begin cooking. A visible timer will count down while cooking (This timer is only visible while in the furnace menu). Overcooking the ingredient will reduce its quality. Ingredients prepared here are considered “Raw”, “Undercooked”, “Cooked” or “Burnt”.
- **Malfunction:**
  If the furnace malfunctions, the flames will flare up, filling the room with smoke and limiting visibility. The player must rapidly press the spacebar to fan the flames and clear the smoke.

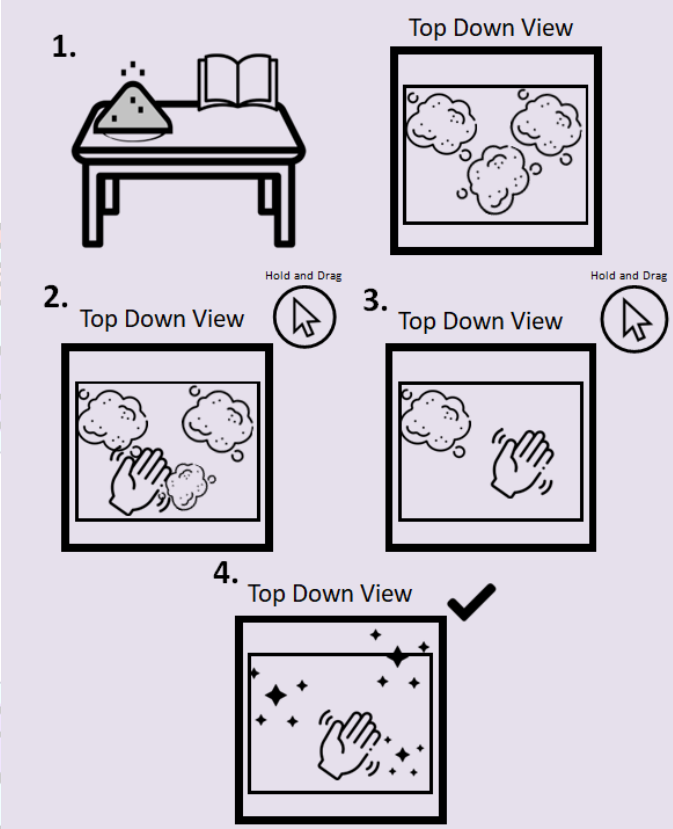


# Enchanter

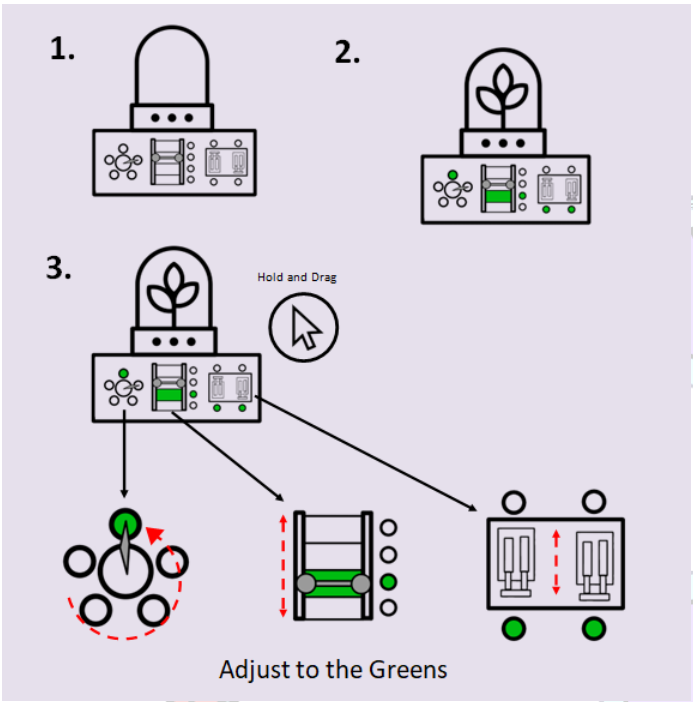
- **Function:**
  You'll be presented with two pages in a book, one containing a full rune symbol and the other with several broken-down rune symbols. Select the correct symbols to complete the rune, gradually increasing its opacity with each correct selection. Once the rune is fully formed, the enchantment is complete. As more time goes on, colors will fade into the symbols to assist players who are struggling.
- **Malfunction:**
  The summoning circle will get covered in ash, and the player must drag the mouse around with left click to clear it away.

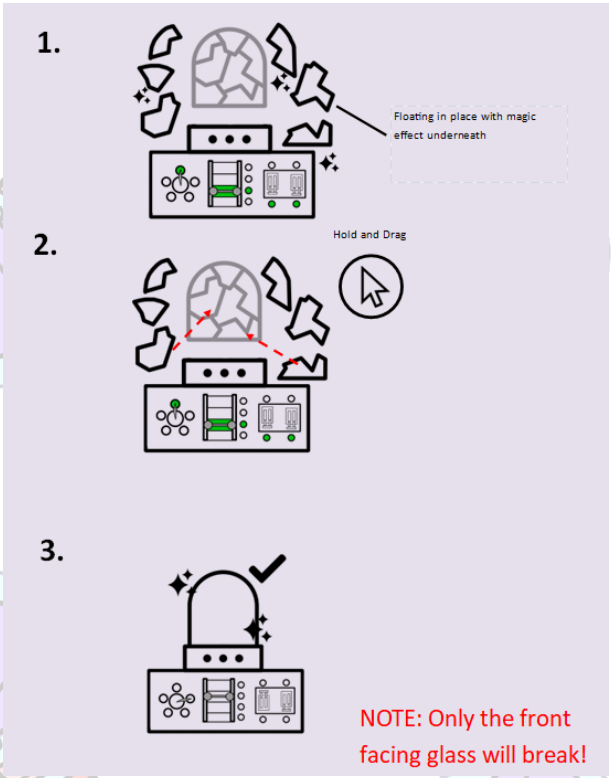


# Incubator

- **Function:**
  When adding an ingredient to the station, different “gems” on the device will light up in blue. Using the mouse, you must adjust the dials and levers on the control panel to face the blue gems as fast as possible. The quality of the incubation depends on how quickly you manage the flashing gems. The difficulty increases based on the complexity of the ingredient being incubated.
- **Malfunction:**
  The incubator will shake violently, causing the glass to shatter. To repair it, the player must solve a jigsaw puzzle, matching the broken glass pieces to the holes in the glass jar. The player will drag the glass pieces by holding down left click on them.



## Work Stations

Cauldron
After an ingredient is prepared, the player can place it in the cauldron to mix it into a potion. The caldron has a tap that allows the player to drain the contents of the cauldron if the want to restart

Flasks
When the player thinks they've placed all the needed ingredients in the cauldron, they can grab a flask and use it with the cauldron. This will drain the cauldron and fill the flask with the potion associated with the ingredients

Bin

Recipe Book
This contains all the available potions and the recipes to make them, along with info on how each craft station works. The books is updated every day as new potions and craft stations are added.

Countertop
Here is where customers show up and order their potions. There is a space at the top of the countertop window that holds a piece of paper which displays the customers current order. At the start of the day, a set of shutters on the countertop are closed. When the player opens these shutters, they'll get their first order and the shit will begin

# Day Cycle

## Overview

Each day has a quota and time limit. The player must sell potions to earn enough money to meet the quota within the time limit, which completes the day and moves onto the next one. Throughout the day, hazards occur at semi-random intervals, depending on the day.

## Hazards

Each day has a certain set of Hazards enabled called the "Hazard List", a counter for when each hazard should occur called the "Hazard Counter", and a "Hazard Intensity" that determines how many hazards are triggered at once. At the start of the day, the Hazard List is shuffled, and the Hazard Counter is set to a value dictated by the day. When a customer arrives, the Hazard Counter decrements by one. When the Counter hits 0, a number of hazards according to the Hazard Intensity are triggered with a random 15-30 second delay between them, and the Counter is reset to the value set by the day. The Hazard List is re-shuffled if all the hazards have been triggered.

## Orders

On each day, there is a list of potions that can be ordered by customers. This list is shuffled into a random order, and when a customer arrives at the counter, they order the next potion in the list. After all of the potions in the list have been ordered, the list is re- shuffled, and the most recently ordered potion is moved to the middle of the list to

avoid consecutive repeats. After a customer orders their potion, the sign on the counter changes to reflect what potion they want.

### Customers

Customers communicate via speech bubbles and icons above their head. The text of the speech bubbles appears one letter at a time with a sound effect. When the player interacts with a customer to give them a potion, they're locked in while the customer pays and the next customer arrives with their order. If the customer receives the wrong potion, they grumble that it's not the one they wanted and leave without paying. If they receive the correct potion, they pay and thank the player according to the star rating of the potion

## Progression

### Overview

Progression is split into multiple days, with craftstations/materials/hazards introduced on certain days. This allows difficulty to grow in a smooth curve, as the player is not overwhelmed by an abundance of complex features. At the beginning of each day, the player receives a letter that briefly explains what features have been added that day, and todays quota

### Day 1

This day serves as a tutorial, where the player is walked through a simple introductory potion

- Materials Unlocked: Plants
- Craftstations Unlocked: Chopping board/ Mortar&Pestle
- New Potions: Drowsiness
- Quota: None

### Day 2

This day adds the Creature material and the Furnace, slightly increasing the complexity of potions. Most potions on this day use Cooked Creatures

- Materials Unlocked: Creatures
- Craftstations Unlocked: Furnace
- New Potions: Health, Love
- Quota: 300

### Day 3

Today introduces the furnace malfunction and potions that require Chopped Creatures.

- Hazards Unlocked: Furnace
- New Potions: Poison
- Quota: 300

## Day 4

Adds the Boiler and Bottled Materials. Potions for this day do not require Boiled Plant ingredients

- Materials Unlocked: Bottled
- New Potions: Night Vision, Scale Skin
- Quota: 300

## Day 5:

Introduces Greater potions and potions that include Boiled Plants

- New Potions: Flight, Greater Health
- Quota: 300

## Day 6

Today adds the Enchanter

- Craftstations Unlocked: Enchanter
- New Potions: Truth, Water Breath
- Quota: 300

## Day 7

Introduces the Boiler and Enchanter malfunctions.

- Hazards Unlocked: Boiler, Enchanter
- New Potions: Growth, Greater Poison
- Quota: 300

## Day 8

This day introduces Minerals. Hazards occur at a reduced frequency on this day, and potions focus on Mineral ingredients

- Materials Unlocked: Minerals
- New Potions: BloodMoon, Strength
- Quota: 300

## Day 9

Today adds the incubator

- Craftstations Unlocked: Incubator
- New Potions: Charm, Blindness, Foresight
- Quota: 300

## Day 10

New potions added

- New Potions: Necromancy, Fragility, Greater Flight
- Quota: 300

# Narrative

## Story

*Background*

**STORY 1**

Our Witch has always dreamed of being the greatest Potion Brewer to ever exist, after completing a master's degree at (UBB) University of Brilliant Brewers, she was well equipped to take on the world! With hope in her eyes and a severe amount of student debt, she took off to the Kingdom to prove her worth. However, only a few weeks into her business startup, the Queen died from what was a suspected poison potion she drank (believing it to be a vital healing elixir). In an outrage the King ordered the criminalisation of potions throughout the Kingdom, believing all brewers to be “shady drug dealers”.

Our humble Witches store was immediately shut down and she was outcasted to the dismay of the locals, before she knew it she was in a situation worse than death: unemployed. Still clinging to her dreams, she has not given up. With the help of another loan and a back alley, she continues to sell potions into the night. However, if she cannot afford the rent of her dingy shop, the not-so-nice landlord with gladly turn her in for a quick buck.

**STORY 2**

Immediately out of university, our friendly Witch was confronted with the crushing reality of a bachelor’s degree in Potion Brewing. There were no jobs! Mass layoffs and the movement towards this thing called “science and medicine” meant the Potion Industry was at an all-time low. Some Kingdoms, including the one she lived in, even criminalised potions for their “Explosive Capabilities”. Unemployed and drowning in debt, our humble Witch did what any desperate student would do, rent a shoddy house in a shady alleyway by a terrible Landlord. In which she can sell potions out the back window.

She refuses to let her dream die, however, if she can’t pay her rent, it may just be the end…

Pending

Our protagonist, a humble witch, spent her days brewing potions and living a modest life, content with her craft despite her financial struggles. One day, a mysterious, shadowy figure approached her with an offer that promised to change everything. He presented her with a recipe for a potion unlike anything she had ever seen or heard of before, offering a fortune in exchange for its successful creation. Eager for the

opportunity, she accepted without hesitation, hoping the deal would lift her out of poverty.
The day of the potion's testing arrived, and the shady figure appeared with an imposing Landlord by his side. As they eagerly watched, the witch presented the potion—only for it to turn out to be nothing more than water. The Landlord, furious, declared that she had failed and now owed him an immense debt. The shady figure, grinning, explained that she was now bound by this failure, forced to pay the noble every day in exchange for her life unless she wanted to face execution.

### *Introduction*
We start at our poor witch's humble store where the noble comes in and demands she pays each day starting day 1 in increments and he will send his personal guard to come collect payments each day.

## Game World
### *Setting and Tone*
Setting - Medieval Fantasy
Tone - Dark Comedy

### *Areas*
Brewing room

## Characters
### *Protagonist*
Stereotypical witch with a staggering amount of debt

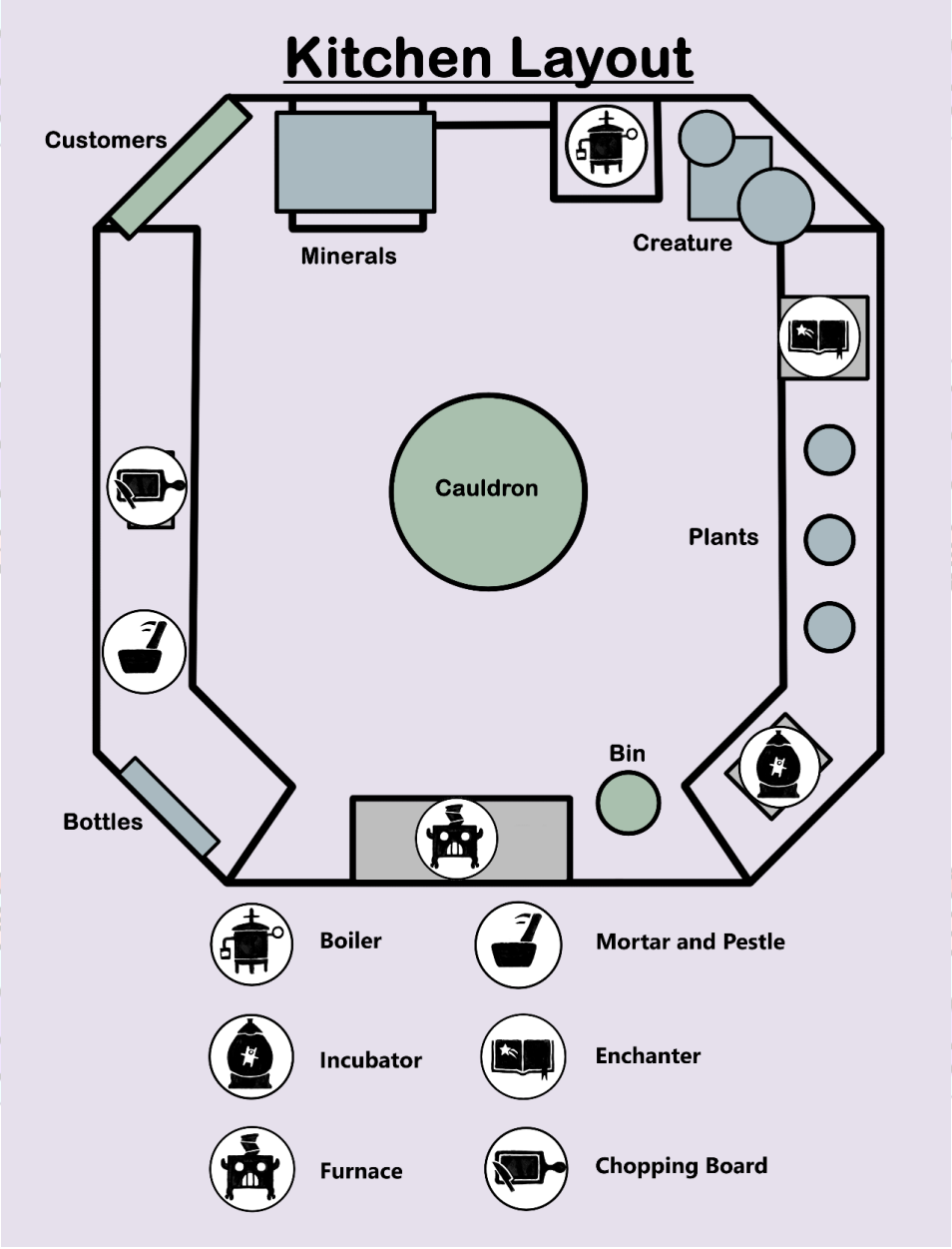
### *Antagonist*
Corrupt Noble

### *Other Characters*
Customers

# Playspace

## Layout

The layout below is design to force players to constantly move forward and back throughout the kitchen in a cramped space. All crafting stations have their



# Technical

## Release Platforms

Potion Commotion will be targeted for release on PC platforms on the Itch.IO website for free download.

## Hardware Requirements

Due to the Low-Poly visuals and singular small scene, the game will be optimised to run on less strenuous hardware. Below are the hardware requirements for our game:

**Minimum Requirements**

**Recommended Requirements**

## Development Hardware & Software

The project will be developed in Unity version 6000.0.19f. Assets and 3D models will be modelled in a mixture of Blender, 3DS Max and ZBrush. Texturing will be done in Substance Painter. Animation and Rigging will be done in both Maya and Motion Builder using Motion Capture for arm movements. All of which will be imported into the Unity Engine.

Additional Software including Microsoft Teams for Communicating, MS Planner for Planning and Miro for brainstorming will be utilised throughout the project.